AAE 558 Finite Element Methods in Aerospace Structures
3 credits

1 Course Information/Description

The goal of AAE 558 is to introduce the theory behind finite element calculations of stress, strain, and deformation in structures and materials, and to describe the role of a commercial finite element package in structural analysis and design. Please note that it's a 5-level course (senior elective and graduate introductory).

2 Learning Resources and Reference Text

Brightspace (BS) Learning Management System:

Access the course videos, materials, and resources through Purdue's Brightspace learning management system. When you log in to Brightspace, start with the Start Here tab, which explains how the course is organized. We strongly recommend that you explore and become familiar not only with the site navigation but also with the content and resources available for this course. Refer to the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:
A First Course in Finite Elements [Paperback] [Jacob Fish (Author), Ted Belytschko (Author) Wiley; 978-0470035801. (available free using Purdue ID online: Go to Purdue Library site→ Search for The Book Title, The free online version is the first search result),

3 Assignments

Your performance in this course will be measured by homework assignments, exams, and a Final Project. Each student will complete a final project. The final project is your final exam. The report should have:

1. Cover page with title, student name (not counted in the 10-page limit)
2. Table of contents page (not counted in the 10-page limit)
3. An abstract page (counted in 10 10-page limit, bullets 3 to 8)
4. Objective or introduction: Describe briefly what the reader will see in your report.
5. Setup or procedure: Describe briefly how you create your FEM model.
6. Describe the model. The number of elements, type of elements, material properties, type of structural member approximation (such as a tapered cross-section being approximated by uniform cross-sections), section properties, boundary conditions, and the load values (you may create a table to present data, take screen snapshots from software, it's up to you).
7. Results and Discussions: Use Figure numbers, Table numbers, and equation numbers in your report to discuss your results. Discuss all the checks (MANDATORY), such as a convergence check, made to ensure your results are satisfactory (all research papers on using FEM are outright rejected without convergence checks). Bring relevant observations to the attention of the reader rather than expecting the reader to wade through the information in your report.
8. Conclusion: Brief synopsis of what you have done in previous steps, with most focus on your findings.

4 Course Learning Objectives

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Objectives AND Envisioned Learning Outcomes:

(1) Understand the relationship between the finite element shape functions, constitutive behavior, and element stiffness matrices
(2) Develop the weak form of the equations of mechanics
(3) Relate mesh and loading to the assembled stiffness matrix
(4) Provide criteria for engineering judgment required to assess the appropriateness of the choice of a finite element model for a particular structure.
(5) Provide training to understand the equations guiding black-box software such as ABAQUS.

5 Evaluation

Grading:

Homework	20%
Class Participation	03%
exam-1 (take-home)	25%
exam-2 (take-home)	25%
preliminary report (take-home)	2%
exam-3 (in the form of take-home final class project)	25%

Grade scale.

$$\begin{bmatrix} A+ & A & A- \\ B+ & B & B- \\ C+ & C & C- \\ D+ & D & D- \end{bmatrix} = \begin{bmatrix} 97\% & 93\% & 90\% \\ 87\% & 83\% & 80\% \\ 77\% & 73\% & 70\% \\ 67\% & 63\% & 60\% \end{bmatrix}$$